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INTRODUCTION

Natural disasters such as earthquakes, floods, hurricanes, wildfires, and landslides pose serious threats to human life, infrastructure, and the environment. In such situations, timely and accurate information is essential for effective disaster response and recovery. Emergency teams and government agencies rely heavily on data to assess the extent of damage, identify affected regions, and allocate resources efficiently.

In recent years, social media platforms like Twitter, Instagram, and Facebook have emerged as important sources of real-time information during disasters. People in affected areas often share images and updates instantly, providing a ground-level view of the situation. However, the large volume of images shared during disasters creates a significant challenge. Not all images are relevant many may be unrelated, duplicated, or of poor quality. Additionally, manually sorting and analyzing thousands of images is time-consuming and impractical.

To address this problem, this project introduces DisasterRes-Net, a multi-stage image classification pipeline designed to automatically analyze disaster-related images.

The proposed system combines deep learning with traditional image processing techniques to improve performance. A pre-trained deep learning model is used to extract high-level features from images, capturing complex patterns such as shapes and structures. In addition to this, handcrafted features such as texture and colour information are also extracted. By combining these different types of features, the system is able to capture both global and local characteristics of the image, leading to more accurate predictions.

The goal of this project is to develop an efficient and scalable solution for analyzing disaster-related images from social media.

LITERATURE REVIEW

The use of technology in disaster management has gained significant attention in recent years, especially with the rapid growth of social media platforms. Researchers have explored different ways to utilize the vast amount of data generated during disasters to improve response and recovery efforts.

Several studies have focused on using social media platforms as a source of real-time information. Unlike traditional data collection methods such as satellite imagery or field surveys, social media allows affected individuals to share updates instantly. This makes it possible to obtain immediate insights into disaster conditions, including infrastructure damage, rescue operations, and affected populations. As a result, many researchers have emphasized the importance of developing automated systems to analyze such data efficiently.

In terms of datasets, multiple publicly available collections have been developed to support disaster-related research.While these datasets have contributed significantly to research, they often have limitations such as class imbalance, limited diversity, or focus on a single task.

Another area of research involves multimodal analysis, where both images and text from social media posts are used together. Some studies have also explored damage localization techniques, which aim to identify specific regions in an image that are affected.

This project builds upon previous research by addressing these limitations. It integrates multiple feature extraction techniques, including deep learning, texture analysis, and color representation, to create a more comprehensive feature set. Furthermore, it uses an ensemble-based classification approach to improve robustness and accuracy. By combining these elements into a single pipeline, the proposed system aims to provide a more effective solution for disaster image analysis.

RESEARCH GAPS

Even though there has been a lot of work done in disaster image analysis, there are still several areas that need improvement. One common issue is that many existing systems focus on only one task at a time. Some models are designed only to identify the type of disaster, while others focus only on estimating damage. In real-life situations, both pieces of information are important, and handling them separately makes the system less practical.

Another problem is that most approaches depend heavily on deep learning models. While these models are powerful, they do not always capture all the details in an image. For example, texture and color patterns can provide useful information, but they are often ignored. Relying only on deep learning can also make the system more complex and require more computational resources.

The quality of data is another important concern. Images collected from social media are often noisy. They may include duplicate images, irrelevant content, or low-quality visuals. Many existing methods do not properly clean or filter this data before training the model, which can reduce accuracy and reliability.

There is also a lack of diversity in the features used for classification. Most models focus only on high-level features extracted from neural networks. This means that important low-level details, such as texture and color distribution, are not fully utilized. As a result, the model may miss useful patterns in the data.

Another limitation is related to the stability of the models. Systems that rely on a single algorithm may perform well on one dataset but fail when tested on different types of disaster images. This shows that many models are not robust enough to handle real-world variations.

Some advanced techniques, such as combining image and text data or detecting specific damaged regions, can improve performance.

However, these methods are often complex and require more processing time, which makes them difficult to use in real-time situations.

To overcome these issues, this project takes a different approach. It combines multiple types of features, including deep learning features, texture-based features, and color information. It also includes proper data cleaning steps to handle noisy inputs. In addition, instead of using a single model, it uses an ensemble method to improve accuracy and stability. Finally, the system is designed to perform both disaster type classification and damage estimation, making it more useful in practical scenarios.

PROPOSED METHODOLOGY AND DATA

The proposed system follows a structured pipeline consisting of multiple stages to process and classify disaster-related images. The methodology can be explained in the following steps:

a) Data Collection
In this step, images are collected from different sources. The primary dataset used is CrisisMMD, which contains disaster-related images. Additional images are collected using web crawling techniques from search engines such as Google and Bing. The collected images are organized into separate folders based on categories like earthquake, flood, hurricane, wildfire, landslide, and non-disaster. These folder names act as labels for classification.

b) Data Cleaning and Preprocessing
The collected dataset is cleaned to improve its quality. Duplicate images are removed using hashing techniques to avoid redundancy. Corrupt or unreadable images are also filtered out. All images are resized to a standard size of 299 × 299 pixels to ensure consistency. After preprocessing, the dataset is split into training and testing sets using a 70 to 30 ratio.

c) Feature Extraction
Multiple types of features are extracted from each image to capture different characteristics. Deep features are obtained using a pre-trained InceptionResNetV2 model, which captures high-level patterns. In addition, texture features are extracted using GLCM and Local Binary Patterns to capture surface details. Color features are also extracted using HSV histograms to represent color distribution.

d) Feature Fusion and Dimensionality Reduction
All extracted features are combined into a single feature vector to improve the representation of the image. Since the combined feature size is large, dimensionality reduction is applied using Principal

Component Analysis. This helps in reducing complexity while preserving important information.

e) Model Training
An ensemble learning approach is used for classification. Multiple models, including Random Forest, Extra Trees, and XGBoost, are combined using a voting mechanism. The model is trained using the training dataset, where features act as input and folder-based labels act as output.

f) Classification Tasks
The system performs two classification tasks. The first task is disaster type classification, where images are categorized into different disaster classes. The second task is damage classification, where images are classified into levels such as little or none, mild, and severe. Both tasks are handled independently using the same feature extraction process.

g) Model Evaluation
The performance of the model is evaluated using the test dataset. Standard evaluation metrics such as accuracy, precision, recall, and F1-score are used. Cross-validation is also performed to ensure that the model is consistent and does not overfit.

The proposed system is designed as a step-by-step pipeline that processes disaster-related images and classifies them based on disaster type and damage level. The overall approach consists of three main stages: data collection, data preprocessing, and feature extraction with classification.

In the first stage, images are collected from different sources. The main dataset used is CrisisMMD, which contains disaster-related images. In addition to this, extra images are gathered using web crawling techniques from search engines. These images are then organized into separate folders based on categories such as earthquake, flood, hurricane, wildfire, landslide, and non-disaster. This folder structure is important because it is later used as the label for classification.

In the second stage, the collected data is cleaned and prepared for training. Duplicate images are removed using hashing techniques to ensure that the dataset does not contain repeated samples. Corrupt or unreadable images are also filtered out. After cleaning, all images are resized to a fixed size of 299 by 299 pixels so that they can be processed consistently by the model. The dataset is then divided into training and testing sets using a 70 to 30 split. This helps in evaluating how well the model performs on unseen data.

The third stage focuses on feature extraction and classification, which is the core part of the system. Instead of relying on a single type of feature, the model extracts multiple types of features from each image. First, deep features are obtained using a pre-trained InceptionResNetV2 model. These features capture high-level information such as shapes and structures present in the image. In addition to this, texture features are extracted using methods like GLCM and Local Binary Patterns, which help in identifying patterns and surface details. Color features are also extracted using HSV histograms to capture color distribution in the image.

All these features are combined into a single feature vector. This process is known as feature fusion, and it helps in improving the overall representation of the image. Since the combined feature vector is large, dimensionality reduction is applied using Principal Component Analysis. This reduces the number of features while retaining most of the important information.

For classification, an ensemble learning approach is used. Instead of a single model, multiple models such as Random Forest, Extra Trees, and XGBoost are combined. These models work together using a voting mechanism to produce the final prediction. This improves accuracy and reduces the chances of overfitting.

The system performs two types of classification tasks. The first task is to identify the type of disaster based on the image. The second task is

to estimate the level of damage, which is categorized into little or none, mild, and severe. These two tasks are handled separately but use the same feature extraction process.

Finally, the performance of the model is evaluated using standard metrics such as accuracy, precision, recall, and F1-score. Cross-validation is also used to ensure that the model performs consistently across different subsets of the data.

Overall, this methodology combines data cleaning, feature fusion, and ensemble learning to create a robust and efficient system for disaster image analysis.

PERFORMANCE MARKERS

To evaluate the effectiveness of the proposed system, several standard performance metrics are used. These metrics help in understanding how well the model performs on unseen data and how accurately it classifies disaster-related images.

Accuracy

Accuracy is calculated as the ratio of correct predictions to the total predictions made by the model. A higher accuracy value indicates better overall performance. However, accuracy alone may not always give a complete picture, especially when the dataset is imbalanced.

Precision

Precision measures how many of the images predicted as a particular class are actually correct. It focuses on the quality of positive predictions made by the model.

High precision indicates that the model makes fewer false positive errors.

Recall

Recall measures how well the model is able to identify all relevant images belonging to a particular class. It indicates the model's ability to capture actual positive cases.

A high recall value means that the model successfully identifies most of the relevant disaster images. This is especially important in disaster scenarios, where missing critical information can affect decision

making.

F1 Score

The F1 score is the harmonic mean of precision and recall. It provides a balance between these two metrics and is particularly useful when there is an uneven class distribution.

A higher F1 score indicates that the model maintains a good balance between precision and recall. In this project, the F1 score is used as an important metric to evaluate overall classification performance.

Cross-Validation Score

To ensure that the model performs consistently, cross-validation is used. The dataset is divided into multiple subsets, and the model is trained and validated on different combinations of these subsets.

The cross-validation F1 score represents the average performance across all folds, while the standard deviation indicates how stable the model is. A low standard deviation means that the model produces consistent results across different data splits.

Confusion Matrix

A confusion matrix is used to visualize the performance of the classification model. It shows the number of correct and incorrect predictions made for each class.

This helps in identifying which classes are being misclassified and provides deeper insight into the strengths and weaknesses of the model.

RESULTS

The proposed DisasterRes-Net system was evaluated on the test dataset for both disaster type classification and damage severity classification. The performance was measured using accuracy, F1-score, and cross-validation results.

Disaster Type Classification

The model was trained to classify images into different disaster categories such as earthquake, flood, hurricane, wildfire, landslide, and non-disaster. The performance of the model is summarized below:

- Accuracy: 73.73%
- F1 Score: 72.64%
- Cross-validation F1 Score: 72.69% ± 0.0097

These results indicate that the model is able to correctly classify most of the disaster types with reasonable accuracy. The F1 score being close to the accuracy shows that the model maintains a good balance between precision and recall. The low standard deviation in cross-validation suggests that the model performs consistently across different data splits.

Damage Severity Classification

The model was also evaluated for classifying the level of damage into categories such as little or none, mild, and severe. The performance results are as follows:

- Accuracy: 76.82%
- F1 Score: 76.39%
- Cross-validation F1 Score: 76.42% ± 0.0108

The results show that the model performs slightly better in damage classification compared to disaster type classification. This may be because damage levels are more visually distinguishable than different disaster types in some cases.

Comparative Analysis

By comparing the two tasks, it can be observed that:

- Damage classification achieves higher accuracy and F1 score than disaster type classification.
- The consistency of both models is high, as indicated by low variation in cross-validation scores.
- The feature fusion approach contributes to stable performance across both tasks.

DISCUSSION

The results show that the DisasterRes-Net system is able to perform both disaster type classification and damage severity classification effectively. The model achieves good accuracy across both tasks, which shows that the overall approach is working well. The combination of different types of features and the use of multiple models together clearly helps improve performance.

One noticeable point is that the model performs slightly better in damage classification than in disaster type classification. This makes sense because damage levels such as severe or mild are usually easier to identify visually. In comparison, different disaster types can sometimes look similar, which makes them harder to distinguish.

The use of feature fusion is one of the main strengths of this system. Deep learning features help capture important patterns in the image, while texture and color features add extra detail. Combining all of these gives the model a better understanding of the image, which leads to more accurate predictions.

The ensemble approach also improves the results. Since multiple models are used together, the final prediction is more stable and less likely to be affected by errors from a single model. This is supported by the consistent cross-validation scores.

Data preprocessing also plays an important role. Cleaning the dataset by removing duplicates and corrupted images improves data quality, which helps the model learn better. Standardizing image size also ensures consistency during training.

Overall, the system performs well and shows that combining different techniques can lead to better results. With more data and further improvements, the model has strong potential to perform even better in real-world applications.

CONCLUSION

In this project, a disaster image classification system called DisasterRes-Net was developed to automatically analyze disaster-related images. The system is designed to perform two main tasks, identifying the type of disaster and estimating the level of damage shown in the image.

The proposed approach combines deep learning with traditional image processing techniques to improve performance. By using multiple types of features, including deep features, texture features, and color features, the system is able to capture different aspects of an image. In addition, the use of an ensemble learning method helps improve accuracy and makes the model more reliable.

The results show that the system is able to achieve good performance in both classification tasks. The model performs consistently across different data splits, which indicates that it generalizes well. The preprocessing steps also play an important role in improving the quality of the dataset and overall model performance.

Overall, the project demonstrates that combining feature fusion and ensemble learning is an effective approach for disaster image analysis. The system can be useful in real-world scenarios by helping disaster management teams quickly understand the situation and make better decisions.

In the future, the model can be further improved by using larger and more diverse datasets, optimizing the model for faster processing, and extending the system to support real-time analysis.

REFERENCES

1. Gupta, T., & Roy, S., "DisasterRes-Net: A framework for analyzing social media images in disaster response," *International Journal of Disaster Risk Reduction*, 2024.
2. Alam, F., et al., "CrisisMMD: Multimodal Twitter Datasets from Natural Disasters," *Proceedings of ICWSM*, 2018.
3. Szegedy, C., et al., "Inception-v4, Inception-ResNet and the Impact of Residual Connections on Learning," *Proceedings of AAAI Conference*, 2017.
4. Haralick, R. M., Shanmugam, K., & Dinstein, I., "Textural Features for Image Classification," *IEEE Transactions on Systems, Man, and Cybernetics*, 1973.
5. Ojala, T., Pietikäinen, M., & Mäenpää, T., "Multiresolution Gray-Scale and Rotation Invariant Texture Classification with Local Binary Patterns," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2002.
6. Pedregosa, F., et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, 2011.
7. Paszke, A., et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," *NeurIPS*, 2019.
8. Chen, T., & Guestrin, C., "XGBoost: A Scalable Tree Boosting System," *Proceedings of KDD*, 2016.